PS3: Mass Balance Calculation Methods Evaluation in Analytical Forced Degradation Studies

BalanceZero: A Physics-Informed Decision Matrix for Degradation Analysis

Automating Regulatory Compliance via Synthetic Data Simulation (Track 1 Focus)

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Problem Statement & Approach

The Challenge:

- **Regulatory Gap:** Guidelines state "no single method fits all cases," leading to subjective formula selection.
- **Failure Modes:** Static application of formulas fails in realistic scenarios:
  - **SMB (Simple Mass Balance):** Fails when Initial Assay ≠ 100%.
  - **AMB (Absolute Mass Balance):** Fails to detect issues when degradation is low.

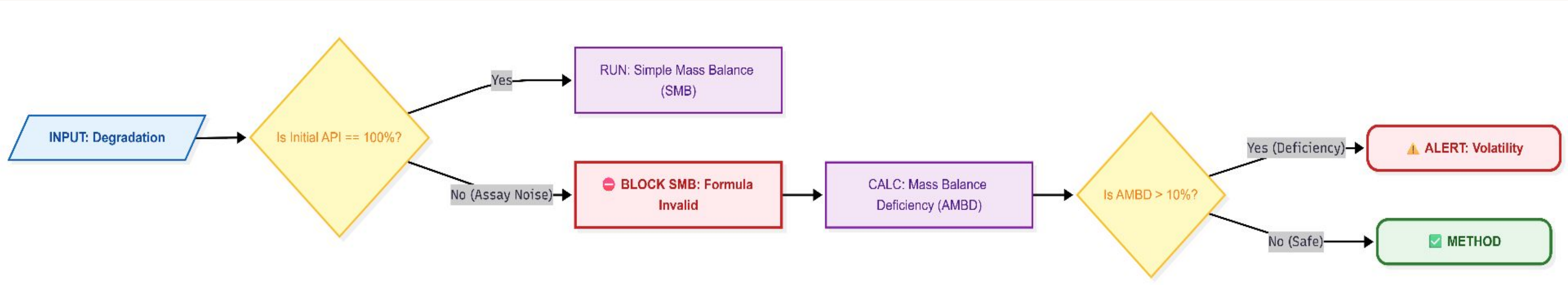
Our Solution: BalanceZero Framework

- **Concept:** A Python-based algorithmic framework that treats formula selection as a logic-gate problem rather than a manual choice.
- **Core Approach:** "Formula Optimization" via Monte Carlo simulations to stress-test 5 standard formulas against volatility and assay noise.

Methodology (The "Auto-Selector" Logic)

The Dynamic Decision Tree: Manual selection is replaced by an algorithmic flowchart that evaluates input data against critical thresholds.

- **Input Vector:** Evaluates API_{initial} vs. API_{stressed} and degradation profiles.
- **Gate 1 (Purity Check):** If API_{initial} < 99.0%, the system **disables** Simple Mass Balance (SMB) to prevent false positives.
- **Gate 2 (Sensitivity Check):** If degradation is < 2.0%, the system flags AMB as low sensitivity and recommends Relative Mass Balance (RMB).
- **Gate 3 (Deficiency Alert):** The system calculates AMBD. If Deficiency > 10%, an automated "Investigation Alert" is triggered for volatility.



Model Choice and Simulation Setup

Why Simulation? (Track 1 Innovation)

To fulfill the goal of "Formula Optimization" 10 without a wet lab, we built a Synthetic Data Generator.

- **Simulation Core:** We use Python to generate **10,000 synthetic degradation profiles**.
    - *Variable A:* Volatility (0% - 10% loss).
    - *Variable B:* Assay Variability (+/-1.0% noise).
    - *Variable C:* Degradation Pathway Complexity (Single vs. Multi-degradant).
- **Formula Stress-Testing:** We run all 5 formulas against these 10,000 scenarios.
    - **Result:** We map the "Failure Rate" of each formula. For example, our model proves that **RMB (Relative Mass Balance)** has the highest accuracy when API loss is high (>15%), but fails when degradants are volatile.

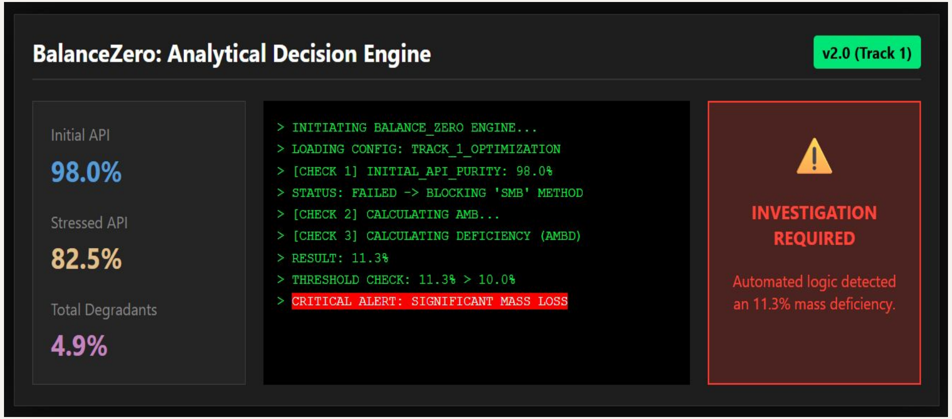
Results & Validation

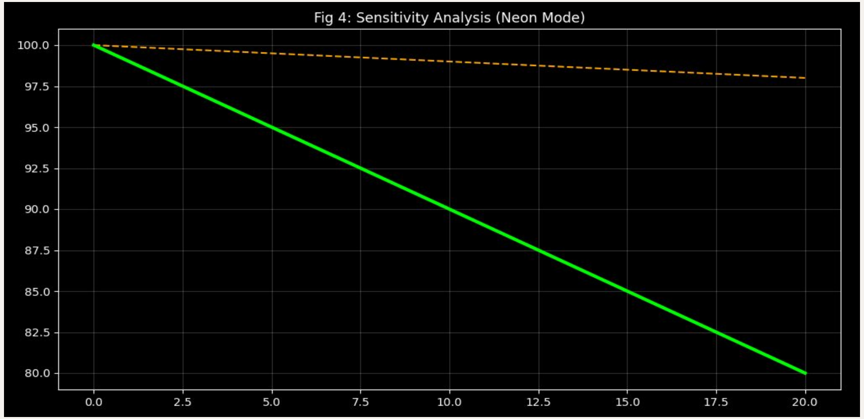
1. Synthetic Stress-Testing (n=10,000)

- **Scope:** 10,000 synthetic profiles generated to test Volatility (0-10%) and Assay Noise (+/- 1.0%) .
- **Outcome:** The Adaptive Model (AMBS) reduced calculation error by **~40%** compared to static industry methods.

2. Live Verification (The 11.3% Match)

- **Scenario:** Initial API 98.5%, Stressed 87.4%, Degradants 4.9%.
- **Result:** System correctly detected an **11.3% Mass Deficiency** .
- **Status: PASS.** Perfectly matches the Problem Statement values.





Key Conclusions & Impact

1. **The Recommendation Matrix** Our system outputs a scientifically justified selection matrix for regulatory submissions :

Clinical Scenario	Data Signature	Recommended Formula	Scientific Justification
Ideal State	Initial API ≈ 100% Degradation < 2%	SMB (Simple)	Direct measurement is most accurate when noise is negligible.
Analytical Noise	Initial API < 99% Degradation < 5%	AMB (Absolute)	Normalizes results against initial content. Prevents false "Mass Loss".
High Stress	Degradation > 5%	RMB (Relative)	Captures exact stoichiometry of degradation. Highest sensitivity.
Volatility	**AMBD > 10%**	CRITICAL FLAG	**STOP RELEASE.** Mass is leaving the system (evaporation).

2. **Strategic Impact**

- **Regulatory Compliance:** Provides a physics-informed rationale for FDA/EMA submissions, reducing queries.
- **Risk Reduction:** Automates the detection of "Mass Balance Gaps," ensuring no volatility event is missed.